

PROGRAMMING FOR PROBLEM SOLVING

II Semester: Common for all Branches										
Course Code		Category		Hours / Week			Credits	Maximum Marks		
A6CS02		ESC		L	T	P	C	CIA	SEE	Total
				3	-	-	3	40	60	100
UNIT-I		INTRODUCTION - PROBLEM SOLVING AND ALGORITHMIC THINKING & INTRODUCTION TO C LANGUAGE								
Algorithm -Definition, Characteristics of Algorithm. Constituents of algorithms: - Sequence, Selection and Repetition. Algorithm with Example: Roots Of a Quadratic Equations, Minimum And Maximum Numbers of a Given Set, Given number is prime number or not, given integer is palindrome or not, etc. Flowchart/Pseudo Code with examples. Introduction To C Language: Structure of C Program, Data Types, data input and output statements, Operators, Precedence and Associativity of operators , Evaluation of Expressions, Type Conversions In Expressions.										
UNIT-II		CONTROL STRUCTURES AND ARRAYS								
Control structures: Decision statements; if and switch statement; Loop control statements: while, for and do while loops, Jump statements: break, continue, goto statements. Arrays: Concepts, One dimensional array, declaration and initialization of one dimensional arrays, two dimensional arrays, initialization and accessing, multi dimensional arrays										
UNIT-III		FUNCTIONS AND POINTERS								
Functions: Function definition, Types of Functions: User defined and built-in Functions, Advantages of User Defined Functions. Parameter passing in functions: Call by value, Call by reference, Passing arrays to functions, Recursion as a different way of solving problems. Example programs, such as Finding Factorial, Fibonacci series, Towers of Hanoi etc.. Storage classes. Pointers: Pointer basics, pointer arithmetic, pointers to pointers, generic pointers, array of pointers, Functions returning pointers, Dynamic memory allocation.										
UNIT-IV		STRINGS AND USER DEFINED DATA TYPES								
Strings: Arrays of characters, variable length character strings, inputting character strings, character library functions, String Handling Functions, Arrays Of Strings Structures and Unions: Structure definition, initialization, accessing structures, nested structures, arrays of structures, Structures and functions, Self-referential structures, unions, typedef, enumerations.										
UNIT-V		FILE HANDLING , SEARCHING AND SORTING								
File Handling: Command Line Arguments, File Modes, Basic File Operations Read, Write And Append, Example Programs. Random Access Using fseek, ftell and rewind Functions. Basic Searching And Sorting Algorithms: Linear and Binary Search, Bubble Sort ,Insertion Sort, Quick Sort.										
Text Books:										

- 1) B.A. Forouzan and R.F. Gilberg C Programming and Data Structures, Cengage Learning, (3rd Edition)
- 2) Byron Gottfried, "Programming with C", Schaum's Outlines Series, McGraw Hill Education, 3rd edition, 2017.
- 3) Programming in C E. Balagurusamy Edition 3 Publisher Tata McGraw-Hill Publishing, 1990

Reference Books:

- 1) W. Kernighan Brian, Dennis M. Ritchie, "The C Programming Language", PHI Learning, 2nd Edition, 1988.
- 2) Yashavant Kanetkar, "Exploring C", BPB Publishers, 2nd Edition, 2003.
- 3) Schildt Herbert, "C: The Complete Reference", Tata McGraw Hill Education, 4th Edition, 2014.4. R. S. Bichkar, "Programming with C", Universities Press, 2nd Edition, 2012.
- 4) Dey Pradeep, Manas Ghosh, "Computer Fundamentals and Programming in C", Oxford University Press, 2nd Edition, 2006.
- 5) Stephen G. Kochan, "Programming in C", Addison-Wesley Professional, 4th Edition, 2014.

Course Outcomes : Students should be able to:

- 1) Apply algorithmic thinking to understand, define and solve problems
- 2) Develop computer programs using programming constructs and control structures and to use arrays to develop C programs
- 3) Decompose a problem into functions to develop modular reusable code and to use pointers to solve complex problems.
- 4) Use Strings and structures to formulate algorithms and programs.
- 5) Use FILE to perform read and write operations.